GEOLOGICAL RESOURCE SCREENING / UK & IRELAND

SILICON & SILICA RESOURCES OF THE BRITISH ISLES

A regional prospectus for identifying quartz, quartzite and silica-sand resources with potential as feedstock for silicon production.



FOCUS AREAS

Scotland · Wales · England · Ireland

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Desktop review - not a mineral resource estimate, feasibility study or investment recommendation.

Executive assessment

Silicon is abundant across the British Isles, but it is generally present as silica (SiO_2) in quartz, quartzite and silica-rich sandstones rather than as naturally occurring elemental silicon. The commercial question is not simply where silica occurs; it is where deposits combine suitable chemistry, grain size, mineralogy, volume and access for the intended conversion route.

KEY FINDING

The UK Critical Minerals Intelligence Centre identifies three quartzite formations for further high-grade silica investigation: Scaraben and Loch Treig in Scotland, and Holyhead in Wales. Limited historical analyses cited in the assessment are around 98-99% SiO_2; the report states that resource potential remains uncertain and that the material is below the greater-than-99.5% silica feedstock benchmark it describes for metallurgical silicon. [1]

Priority locations at a glance

AREA / LOCATION	MATERIAL / EVIDENCE	ASSESSMENT FOR SILICON
Scaraben, Caithness / Sutherland	Quartzite formation; limited analyses reported at about 98-99% SiO_2.	Priority for modern sampling; resource size and impurity profile unresolved. [1]
Loch Treig, Highlands	Quartzite formation; identified by UKCMIC as a potential high-grade silica target.	Priority for modern sampling and beneficiation tests. [1]
Holyhead, Anglesey	Quartzite formation; limited analyses reported at about 98-99% SiO_2.	Priority for modern sampling; no modern resource estimate established here. [1]
Lochaline, Morvern	Very pure quartz sandstone and established silica-sand mine. A 2022 plant-fines sample analysed at 99.1 wt.% SiO_2.	Strong industrial silica reference point; sand grain size may not suit conventional silicon-metal furnaces. [2]
Dalwhinnie, Highlands	Historic vein-quartz occurrence investigated for industrial use; reported resources thought small.	Secondary exploration lead, not a demonstrated large-scale source. [1]
Carrickleck, County Meath	Disaggregated Upper Carboniferous sandstone; quarry documented in Geological Survey of Ireland heritage audit.	Known Irish silica-sand occurrence; chemistry, current status and scale need confirmation. [6]

A separate industrial-sand network

Commercial silica sand is also produced or historically worked in several English regions, including Cheshire, Norfolk, Surrey, Kent, Bedfordshire, Nottinghamshire and North Yorkshire, as well as parts of Scotland. These are important industrial mineral occurrences, but should not automatically be treated as quartz feedstock for silicon-metal furnaces. The BGS notes that many UK silica sands and weakly consolidated sandstones are too fine-grained for standard silicon-metal processing. [1, 4]

HOW TO READ THIS REPORT

The report separates (1) the existence of silica, (2) suitability as silicon-metal furnace feedstock, and (3) suitability for the much more demanding purification chain needed for semiconductor silicon. These are different technical and commercial tests.

Scotland: the principal quartzite targets

Scotland has the most prominent named quartzite prospects in the UKCMIC screening, alongside an established high-purity silica-sand operation at Lochaline. The geological targets are not interchangeable: quartzite is a consolidated rock that may provide coarse furnace feed, whereas Lochaline is a quartz sandstone mined and processed as industrial silica sand.

1. Scaraben Quartzite - Caithness / Sutherland

The Scaraben Quartzite is a named geological unit in the far north of Scotland. UKCMIC lists Scaraben among the three UK quartzite formations with potential for high-grade silica applications. Limited analyses referenced in that assessment are around 98-99% SiO_2, but the report cautions that the resource potential is uncertain. [1]

- Work programme: map the formation and structural continuity; collect representative channel and core samples; determine silica grade and the full trace-element profile.
- Main uncertainties: aluminium-bearing minerals, iron oxides, accessory minerals, weathering variability, recoverable tonnage and practical access.

2. Loch Treig Quartzite - Scottish Highlands

Loch Treig is the second Scottish quartzite target specifically highlighted by UKCMIC. It warrants a modern assessment of grade continuity, impurities and physical characteristics. The presence of quartzite in the wider Highlands does not itself establish a commercially usable deposit; the BGS assessment notes that many UK quartzites have lower silica and higher aluminium contents. [1]

- Work programme: distinguish the target quartzite unit from adjacent lithologies; test across strike and down dip; establish crushing behaviour, lump yield and fines generation.

3. Lochaline - Morvern Peninsula

Lochaline White Sandstone is described by the BGS Lexicon as a very pure, white to pale yellow-brown, well-sorted, medium-grained quartz sandstone. [3] A 2024 BGS/NERC mineral-waste factsheet reports that a sample of hydrosizer fines collected at Lochaline on 4 October 2022 contained 99.1 wt.% quartz by X-ray diffraction and 99.1 wt.% SiO_2 by chemical analysis. The same factsheet describes the material as the purest silica sand in the UK and lists commercial products differentiated by iron content, including LQS85 and LQS500 specifications. [2]

IMPORTANT PROCESSING DISTINCTION

The 99.1% figure belongs to a specified sample of plant hydrosizer fines; it is not a blanket assay for every part of the deposit or a certificate for semiconductor use. Lochaline's sand is valuable industrial feedstock, but UKCMIC states that many fine silica sands are unsuitable for standard silicon-metal furnace processing because of their particle size. [1, 2]

4. Dalwhinnie and other vein quartz occurrences

UKCMIC notes that vein quartz is common in the UK but that no deposit of sufficient size to merit consideration as a silicon-metal silica source had been identified in the assessment. A vein near Dalwhinnie was historically worked or investigated for industrial applications, but its resources were considered small. It is best treated as a secondary lead for high-purity sampling rather than a proven bulk source. [1]

5. Kinlochleven - Glencoe and wider Highland quartzites

The wider Highlands contain multiple quartzite-bearing successions. These may be useful for regional prospectivity mapping, but the existence of quartzite alone is not evidence of metallurgical suitability. Any local target outside the named UKCMIC prospects should be screened for aluminium, iron, boron, phosphorus and accessory mineral content before it is advanced.

Wales, England and Ireland

Wales: Holyhead Quartzite

The Holyhead quartzite formation on Anglesey is the Welsh formation identified by UKCMIC as having potential for high-grade silica applications. Limited analyses cited in the assessment are around 98-99% SiO_2, and the resource potential is explicitly described as uncertain. A modern programme should establish whether suitable low-impurity material occurs consistently over a mineable volume, rather than relying on isolated historical samples. [1]

- Key checks: grade continuity, aluminium-bearing phases, iron staining, structural breaks, overburden, land access and environmental constraints.

England: established silica-sand districts

England has a broader industrial silica-sand footprint than the UK quartzite targets. BGS and planning material identify important producing or resource areas in Cheshire, Norfolk, Surrey, Kent, Bedfordshire, Nottinghamshire and North Yorkshire. Other historically identified areas include Essex and Worcestershire. [4, 5]

DISTRICT	GEOLOGICAL / INDUSTRIAL CONTEXT	WHAT IT MAY OFFER
Cheshire	Pleistocene silica sands	Established industrial silica supply; test for fines and contaminants before any metallurgical route. [4,5]
Norfolk	Cretaceous silica-sand resources	Industrial glass and foundry-sand resource district. [4,5]
Surrey and Kent	Lower Cretaceous / Folkestone-type sand resources	Industrial silica occurrences; site-specific chemistry and permissions matter. [4,5]
Bedfordshire	Woburn Sands; Leighton Buzzard district	Long-established industrial sand workings; not equivalent to coarse quartzite. [7]
Nottinghamshire / North Yorkshire	Other identified industrial silica districts	Potential supplementary feedstock research locations; deposit-by-deposit screening required. [5]

Ireland: Carrickleck, County Meath

The Geological Survey of Ireland's Meath geological-heritage audit documents Carrickleck Quarry, near Nobber, in the Carrickleck Sandstone Member of Upper Carboniferous age. The report describes locally disaggregated sandstone that can be crushed by hand to produce fine sand, and records quarry faces reaching up to about 30 metres. The audit is historical, so current ownership, operating status, reserves and product chemistry must be independently checked. [6]

INTERPRETATION

Carrickleck is evidence of a documented Irish silica-sand occurrence, not proof of a semiconductor-grade quartz resource. Ireland-wide prospectivity should be explored through Geological Survey Ireland mapping and quarry records, then tested using the same assay and process criteria as UK targets.

From geology to semiconductor feedstock

A deposit can be valuable for glass, foundry or filtration markets and still be unsuitable for silicon-metal manufacture. Equally, a silica source that can be reduced to metallurgical-grade silicon is still several purification stages away from semiconductor-grade material.

1 GEOLOGICAL RESOURCE	2 BENEFICIATION	3 SILICON METAL	4 CHEMICAL PURIFICATION	5 CRYSTAL / DEVICE
Quartzite, vein quartz or silica sand; establish grade, volume and contaminants.	Crushing, sizing, washing, sorting and impurity reduction; retain suitable grain size.	Carbothermic reduction in a high-temperature furnace to metallurgical-grade silicon.	Convert to volatile silicon compounds, purify by chemical separation and deposit high-purity silicon.	Crystal growth, wafer preparation, doping and semiconductor fabrication.

What makes a silica deposit suitable?

For conventional silicon-metal production, UKCMIC describes feedstock as pure silica or quartz with more than 99.5% SiO_2 and a low fines content. The same assessment notes that UK quartzites often have lower silica and higher aluminium, and that the three named prospects need further evaluation. This is a screening benchmark, not a universal purchase specification: furnace operators will also specify particle size, strength, permeability, thermal behaviour and impurity limits. [1]

ANALYTE / PROPERTY	WHY IT MATTERS	TESTING APPROACH
SiO_2 and major oxides	Establish bulk silica grade and main gangue components, especially Al_2O_3, Fe_2O_3, CaO and MgO.	XRF on representative prepared samples; confirm by wet chemistry where required.
Boron and phosphorus	Important trace contaminants for downstream silicon purification and electrical performance.	Trace-element methods with suitably low detection limits, selected with the intended process route in mind.
Ti, Na, K and other trace elements	May affect processing, product specification or chemical purification costs.	ICP-MS / ICP-OES and targeted trace-element analysis.
Mineralogy and inclusions	Identifies mica, feldspar, iron oxides, heavy minerals and locked contaminants.	XRD, petrography and SEM-based mineral mapping.
Particle size and strength	Furnace feed requires appropriate size distribution, low fines and robust lump material.	Sieve analysis, attrition/crush tests and process trials.

SEMICONDUCTOR-GRADE IS NOT A MINE-GATE GRADE

A high SiO_2 assay does not make a mineral deposit semiconductor-grade. The silica must first be converted into silicon, then purified through an appropriate chemical route. Semiconductor use requires exceptionally low impurity concentrations and tight control over the full process chain; ore qualification alone cannot certify chip suitability.

Recommended British exploration programme

A staged programme can avoid spending heavily on a deposit before its chemistry, geometry and processing behaviour are understood.

Stage 1 - desktop screening and access

- Compile BGS GeoIndex, geological maps, borehole logs, mineral assessment reports, historic quarry records and relevant Geological Survey Ireland mapping.
- Define target formation boundaries and separate mapped outcrop from inferred subsurface extent.
- Check land ownership, mineral rights, planning designations, protected habitats, water constraints, transport routes and community impacts.

Stage 2 - representative field sampling

- Use mapped traverses and channel samples across lithological changes; use core drilling only where the desktop case justifies it.
- Record exact sample coordinates, elevation, lithology, alteration, weathering, structure, sample mass and chain of custody.
- Collect duplicates, blanks and certified reference materials; avoid basing decisions on attractive hand specimens or isolated high assays.

Stage 3 - laboratory qualification

- Run major-oxide XRF, trace-element assays, XRD mineralogy, petrography and SEM-based inclusion analysis.
- Pay particular attention to Fe, Al, Ti, Ca, Mg, B, P, Na and K, with analytical detection limits set by the future process route.
- Measure grain-size distribution, fines generation, crushing strength and the response to washing, magnetic separation, flotation or other beneficiation tests.

Stage 4 - process and economic tests

- Prepare representative bulk samples and test whether beneficiation can produce consistent, low-contaminant feed at acceptable mass yield.
- For quartzite, assess whether a commercially useful coarse fraction can be made without excessive fines or energy-intensive crushing.
- For silica sands, assess the economics of cleaning and high-value industrial markets separately from the silicon-metal route.
- Only after successful qualification should furnace testwork, pilot-scale conversion and downstream silicon purification studies be commissioned.

SUGGESTED INITIAL TARGET PORTFOLIO

Begin with Scaraben, Loch Treig and Holyhead as the specific quartzite targets identified by UKCMIC. In parallel, use Lochaline as a benchmark for high-purity industrial silica and evaluate whether selected processing residues or off-spec fractions have value in separate applications. Treat Dalwhinnie and wider Highland quartzites as secondary leads pending evidence of scale and chemistry.

Key data gaps before investment

No modern, independently verified resource estimates, complete impurity datasets, mining-cost models or silicon-furnace test results are presented here for the named quartzite targets. The available 98-99% SiO_2 figures are limited historical analyses cited by UKCMIC, not current resource-wide grades. Lochaline's 99.1% result is one specifically identified hydrosizer-fines sample, not a deposit-wide average. Carrickleck's cited audit is historical and does not establish present operating status or current reserves. [1, 2, 6]

Conclusions and source register

The British Isles have abundant silica-bearing geology and a mature industrial silica-sand sector. The more demanding question for domestic silicon-metal production is whether a sufficiently large, consistent, low-impurity quartz or quartzite body can be established with the right particle characteristics. Current public screening points to Scaraben, Loch Treig and Holyhead as the clearest named quartzite targets, while Lochaline is a strong industrial-silica benchmark. None should be described as a proven semiconductor feedstock without modern sampling, process tests and downstream qualification.

Practical next step

Commission a comparative sampling and assay campaign across the three named quartzite formations, with Lochaline included as a benchmark material. The deliverable should be a georeferenced dataset of assays, mineralogy, particle-size distributions, sample descriptions and beneficiation yields. That would turn a regional geological hypothesis into a decision-grade exploration screen.

SOURCES AND FURTHER TECHNICAL READING

[1] UK Critical Minerals Intelligence Centre / UKCMIC. Potential for Critical Raw Material Prospectivity in the UK, report CR23024, section 7.10, Silicon. Includes the >99.5% SiO_2 feedstock discussion, particle-size caveat, Dalwhinnie vein quartz note and Scaraben, Loch Treig and Holyhead assessments. Open report (PDF).

[2] British Geological Survey / NERC Open Research Archive. UK Mineral Waste Factsheets, report OR24040, Lochaline Mine entry. Records the 4 October 2022 hydrosizer-fines sample, 99.1 wt.% quartz / 99.1 wt.% SiO_2 analysis and product iron specifications. Open report (PDF).

[3] British Geological Survey. Lexicon of Named Rock Units: Lochaline White Sandstone Formation. Geological description of the well-sorted quartz sandstone. Open BGS Lexicon entry.

[4] British Geological Survey. Industrial Mineral Assessment Unit reports and maps. BGS index to regional sand, gravel and industrial mineral assessments; useful for locating historic survey maps and reports. Open BGS resource page.

[5] Surrey County Council. National Planning Framework / silica sand evidence paper, sections on UK silica sand production and distribution. Summarises the principal silica-sand districts in Britain and historic production geography. Open report (PDF).

[6] Geological Survey of Ireland. The Geological Heritage of Meath: County Geological Site Report - Carrickleck Quarry. Documents the Carrickleck Sandstone Member and quarry geology. The report is historical; current status should be checked. Open report (PDF).

[7] British Geological Survey. Geology of the Leighton Buzzard-Ampthill district. Describes Woburn Sands workings and their industrial mineral context. Open BGS memoir.

Additional mapping: BGS MineralsUK and Geological Survey Ireland are starting points for current mapping, mineral information and datasets.

SCOPE AND LIMITATIONS

This is a desktop screening report compiled from public geological and mineral-industry sources available by 02 November 2026. It is not a mineral resource or reserve statement under any reporting code, does not establish economic viability, and does not replace site-specific geological investigation, laboratory testing, permitting review or engineering studies. Locations are regional targets, not recommendations to enter or sample private or restricted land.

Prepared 02 November 2026 | Units: SiO_2 values are stated as reported by the cited sources; wt.% means weight percent.